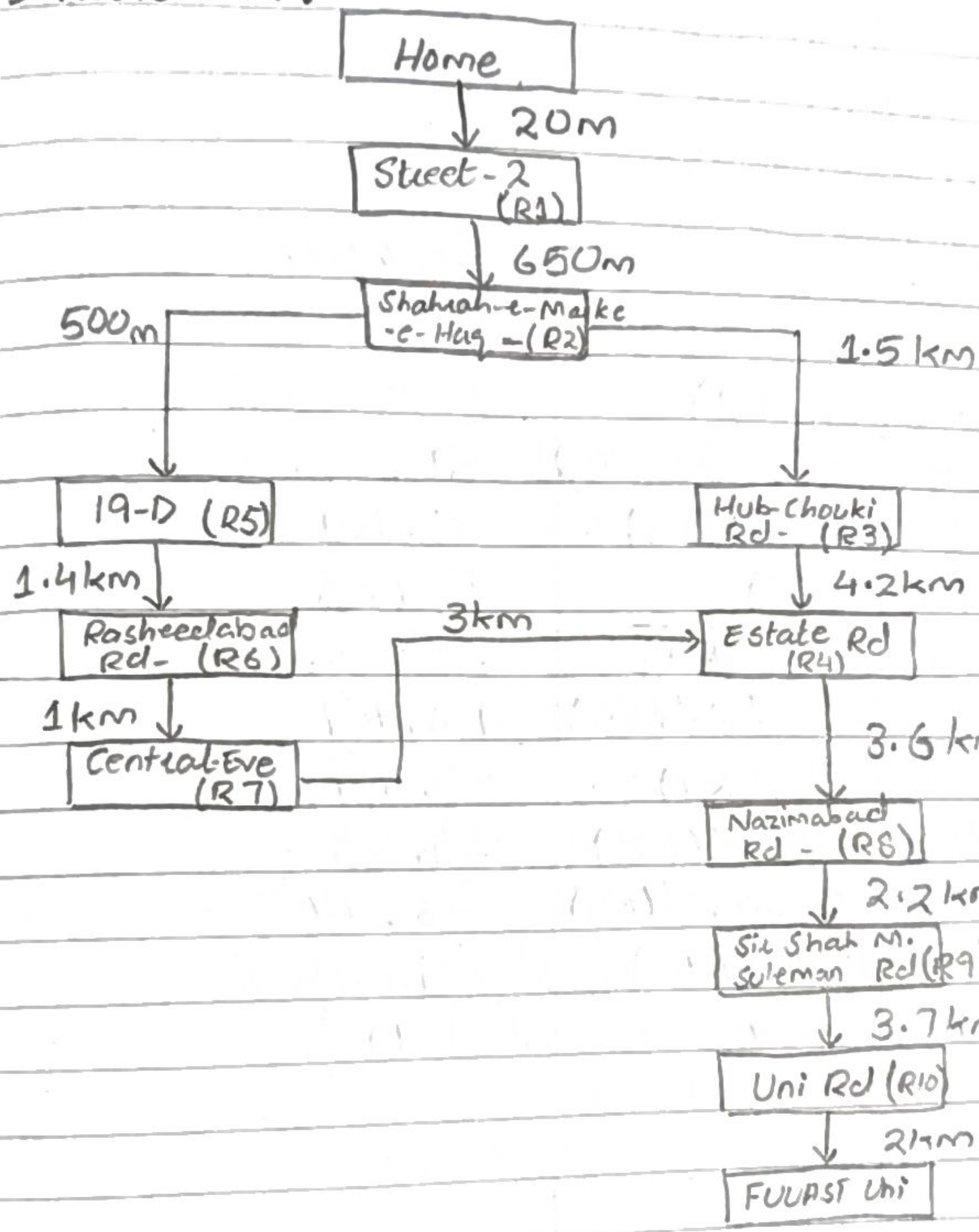
HRISHA - 23122106 - 6A

HOME TO UNI ROUTE

BFS

1. ROUTE MAP:



Route 1:

0.02 + 0.65 + 1.5 + 4.2 + 3.6 + 2.2 + 3.7

+ 2 = 17.87 km

Home ⟶ Street 2 (R_1)

R_1 ⟶ Shahrh-e- Marke Haq, (R_2), Home

R_2 ⟶ 19+D(R_5), Hub Chouki Rd (R_3), R_1

R_5 ⟶ Rasheedabad Rd (R_6), R_2

R_3 ⟶ Estate Rd (R_4), R_2

R_6 ⟶ Central Eve (R_7), R_5

R_4 ⟶ Nazimabad Rd (R_8), R_3

R_7 ⟶ R_6, R_4

R_8 ⟶ Sir Shah m. Suleman Rd (R_9), R_4

R_9 ⟶ Uni Rd (R_{10}), R_8

R_{10} ⟶ Fuvast Uni, R_9

Fuvast Uni ⟶ R_{10}

Roads	h(n)
	17.876m
Home	17.85
R_1	17.2
R_2	17.37
R_5	15.70
R_3	15.97
R_6	11.50
R_4	14.97
R_7	7.90
R_8	5.70
R_9	2.00
R_{10}	0
Furuest Unit	

Step 0 (Initialization)

PQ = Home
V = ∅
f(n) = 17.87km

Parent Table:
Home → None

Step 1: Expand Home

Neighbour = R_1
PQ = [R_1]
V = [Home]

Node	g	h	f
R_1	1	17.87	18.87

Choose R_1 (f = 18.87)

Parent Table.
Home → None
R_1 → Home

Step 2: Expand R_1

Neighbour = R_2, Home(Already visited)
PQ = R_2
V = Home, R_1,

Node	g	h	f
R_2	2	17.2	19.2

Choose R_2 (f = 19.2)

Parent Table
Home → None
R_1 → Home
R_2 → R_1

Step 3 Expand R_2

Neighbour = R_5, R_3, R_1
PQ = R_5, R_3
V = Home, R_1, R_2

Node	g	h	f
R_5	3	17.37	20.37
R_3	3	15.70	18.70

Choose R_3 (f = 18.70)

Parent Table

Home → None
R_1 → Home
R_2 → R_1
R_3 → R_2

Step 4: Expand R_3

Neighbour = R_4, R_2
PQ = R_5, R_4
V = Home, R_1, R_2, R_3,

Node	g	h	f
R_4	4	11.50	15.50

Choose R_4 ($f = 15.50$)

Parent Table:

Home → None
R_1 → Home
R_2 → R_1
R_3 → R_2
R_4 → R_3

Step 5:

Expand R_4

Neighbour = R_8, R_3, R_7
PQ = R_5, R_8, R_7
V = Home, R_1, R_2, R_3, R_4

Node	g	h	f
R_8	5	7.90	12.90
R_7	5	14.97	19.97

Choose R_8 ($f = 12.90$)

Parent Table

Home → None
R_1 → Home
R_2 → R_1
R_3 → R_2
R_4 → R_3
R_8 → R_4

Step 6: Expand R_8

Neighbour = R_9, R_4
PQ = R_5, R_7, R_9
V = Home, R_1, R_2, R_3, R_4, R_8

Node	g	h	f
R9	6	5.70	11.70

Choose R9 (f = 11.70)

Parent Table

Home → None
R1 → Home
R2 → R1
R3 → R2
R4 → R3
R8 → R4
R9 → R8

Step 7: Expand R9

Neighbour - R10, R8
PQ = R5, R7, R10
V = Home, R1, R2, R3, R4, R8, R9

Node	g	h	f
R10	7	5.70	12.70

Choose R10 (f = 12.70)

Parent Table

Home → None
R1 → Home
R2 → R1
R3 → R2
R4 → R3
R8 → R4
R9 → R8
R10 → R9

Step 8: Expand R10

Neighbour = Fuuast Uni, R9
PQ = R5, R7, Fuuast Uni
V = Home, R1, R2, R3, R4, R8, R9, R10

Node	g	h	f
Fuuast Uni	8	0	8

Choose Fuuast Uni (f = 8)

Parent Table

Home → None
R1 → Home
R2 → R1
R3 → R2
R4 → R3
R8 → R4
R9 → R8
R10 → R9
Fuuast Uni → R10